Lecture 5
Developing a RAG-Based Chatbot Research Project

problem • objectives • RAG design • experiment • evaluation

ITC-C508 – Deep Learning (NLP)
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Learning Outcomes

- Identify a real, manageable problem that can be addressed by a RAG-based chatbot.
- Select and justify a relevant UN Sustainable Development Goal (SDG).
- Formulate aligned research objectives and research questions.
- Describe the basic RAG pipeline and knowledge-base preparation.
- Design a simple experiment and evaluation plan.
- Explain what makes the project research—not just software development.

1. Start With a Real Problem

Good research begins with a real need—not just a technology idea.

Example: **"I want to build a RAG chatbot."**

WEAK | Technology-First

- Focuses on the technology, not the user need
- Does not identify who has the problem
- Does not explain why the problem matters

SHIFT from technology to need

Example: **"Users struggle to find reliable answers in many documents."**

BETTER | Problem-First

- Identifies a real user need
- Provides context for the problem
- Can be investigated and measured
- Can justify RAG as a possible solution

THEN ASK

Can RAG make this information easier to find and use?

The technology should follow the problem—not the other way around. A clear problem gives the research purpose, direction, and measurable focus.

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2. Select a UN Sustainable Development Goal (SDG)

Use the SDG to establish why your problem matters.

1. **Choose an SDG** — Select one SDG directly connected to your intended problem.
2. **Identify a related subdomain** — Narrow the SDG to the area relevant to your topic.
3. **Identify a real issue** — Find a real-world issue that affects people or organizations.
4. **Define the specific problem** — State the problem that your chatbot could realistically address.
5. **Identify the RAG contribution** — Explain how the chatbot could help address the problem and support the SDG.

Start with the problem you care about: use the SDG to show its broader relevance.

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3. Match the Problem to an SDG

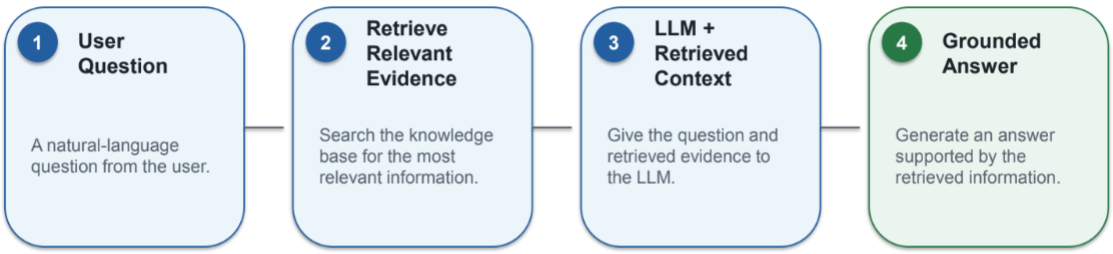
The best SDG is the one that is directly supported by the problem—not simply the most attractive one.

Example project	Possible SDG alignment
University information assistant	SDG 4 – Quality Education
Disaster-preparedness assistant	SDG 11 – Sustainable Cities and Communities
Public-service information assistant	SDG 16 – Peace, Justice and Strong Institutions
Health-information assistant	SDG 3 – Good Health and Well-Being

Choose the SDG that is directly supported by your problem. The SDG provides context and relevance—it does not prove that the chatbot is effective.

4. What Is RAG?

RAG retrieves relevant information from your knowledge base before the LLM generates the answer.



Think of RAG as an open-book exam: find the relevant page first, then answer.

Important: RAG can reduce unsupported answers, but retrieval and generation can still fail.

5. The RAG Pipeline

Separate what happens when the knowledge base is prepared from what happens when a user asks a question.

A. KNOWLEDGE BASE PREPARATION (offline / when documents are updated)

1. Documents → 2. Chunks → 3. Embeddings → 4. Vector Store

Prepare and index the knowledge so it can be searched by meaning.

B. QUESTION ANSWERING (real time)

5. User Question → 6. Retrieve Top-K → 7. LLM + Context → 8. Grounded Answer

Key idea: retrieve the evidence first; then generate the answer using that evidence.

6. Research Methodology: What Will You Actually Do?

A good proposal shows both how the system will be built and how it will be tested.

1. Define the problem
2. Prepare the knowledge base
3. Develop the RAG system
4. Run the experiment
5. Evaluate performance
6. Analyze and explain results

SYSTEM DEVELOPMENT

- Collect and prepare documents
- Build the retrieval pipeline
- Connect the LLM
- Create the chatbot interface

RESEARCH EVALUATION

- Define the test questions
- Compare the selected configurations
- Measure retrieval and answer quality
- Analyze and explain the results

Remember: building the chatbot is the development task; testing whether it works is the research task.

7. What Is Your Dataset in a RAG Project?

In RAG, the main data may be a document collection rather than a labeled table.

A. KNOWLEDGE BASE

Policies	rules, procedures
Manuals	how-to information
FAQs	common questions
Reports	domain information
Approved webpages	current information

B. TEST QUESTION SET

- Realistic questions users might ask
- Questions whose answers are in the knowledge base
- A few questions whose answers are not available
- Expected answers or supporting passages for checking results

Document source + quantity + date + language + permission status should be recorded.

Prepare the test questions independently from the chatbot being evaluated.

8. Turn the Problem Into Research Objectives

Your objectives should say what you will develop and what you will investigate.

GENERAL OBJECTIVE

To develop and evaluate a RAG-based chatbot that answers domain-specific questions using a curated knowledge base.

SPECIFIC OBJECTIVES

1. **Prepare** a domain-specific knowledge base.
2. **Develop** a RAG pipeline for retrieval and answer generation.
3. **Evaluate** retrieval performance and answer quality.
4. **Compare** two reasonable configurations or a RAG vs. non-RAG baseline.

A strong objective links: PROBLEM → DEVELOPMENT → EVALUATION.

9. Research Questions, Variables, and Evidence Must Align

Research Question	What You Vary / Compare	What You Measure
Does retrieval find the needed passage?	Top-K / retrieval setup	Hit@K / retrieval success
Are answers correct and relevant?	RAG vs. baseline / config	Correctness / relevance
Are answers supported by evidence?	Prompt or RAG setup	Groundedness / faithfulness

A research question should have a method for collecting evidence.

10. Choose a Simple, Fair Experiment

Test at least two design choices. Keep the other conditions the same. But more design choices will earn additional points

Possible design choices

1. **Chunk size**
   Small (e.g., 300 tokens) vs. medium (e.g., 500 tokens).
2. **Top-K retrieved chunks**
   Compare Top-K = 3 vs. Top-K = 5.
3. **Embedding model**
   Compare two embedding models.
4. **RAG vs. non-RAG**
   Compare RAG with a standard LLM without retrieval.

Keep these constant

- Same knowledge-base documents
- Same test questions
- Same LLM and prompt
- Same evaluation method and metrics
- Same temperature / key settings

Tip: A focused experiment is easier to interpret than testing many factors at once.

Change only ONE or TWO factors so you can clearly identify what makes a difference.

11. How Will You Know It Worked?

Use clear evaluation criteria that turn system behavior into measurable evidence.

Evaluation aspect	What it means	Example measure
Retrieval performance	Was the needed supporting passage found?	Hit@K or Recall@K
Answer correctness	Is the answer factually right?	Human rating or reference comparison
Answer relevance	Does the answer actually address the question?	Human rating
Groundedness	Is the answer supported by retrieved evidence?	Evidence-supported rating
User evaluation	Is the chatbot useful and easy to use?	Survey: usefulness, ease, satisfaction

Use automatic metrics where appropriate, but do not rely on a single metric to judge the chatbot.

12. Build a Small but Useful Test Set

A good test set helps you evaluate the chatbot fairly and consistently.

Recommended structure

1. **Prepare 30–50 realistic questions**
   A practical starting point for a senior undergraduate project.
2. **Include different difficulty levels**
   Easy, moderate, and difficult questions.
3. **Cover in-scope and out-of-scope cases**
   Some answers should be in the knowledge base; a few should not be.
4. **Prepare reference answers or passages**
   Use these to check correctness and groundedness.
5. **Create the test set independently**
   Do not let the chatbot generate its own evaluation questions.

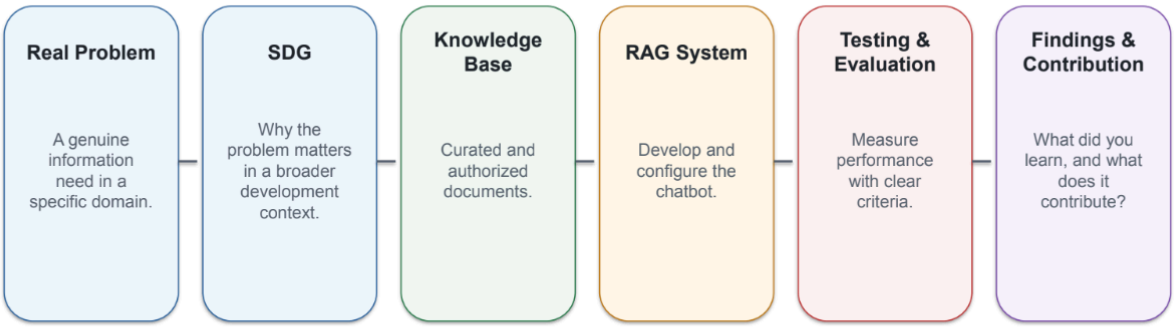
Example question mix

Easy / In-scope	"What are the university office hours?"
Moderate / In-scope	"What documents are required for course transfer?"
Difficult / In-scope	"According to the 2023 policy, what are the grounds for academic probation?"
Out-of-scope	"Who is the current mayor of Manila?"

A well-prepared test set is the basis for fair and meaningful evaluation.

13. Your Proposal Should Tell One Clear Story

Connect the problem, solution, experiment, and contribution.



Problem → Justified Solution → Development → Experiment → Results

That is what turns "we built a chatbot" into an IT research project.